

Treasury return predictability and investor sentiment

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Abstract

We document that the Treasury market investor sentiment (TSENT) of institutional investors is a powerful predictor of bond risk premia. Specifically, TSENT positively predicts Treasury bond excess returns in and out of sample. The forecasting gains of TSENT are incremental to those in conventional bond return predictors: Fama–Bliss forward spreads, Cochrane–Piazzesi forward rate factor, and Ludvigson–Ng macro factor, as well as equity market sentiment proxies such as the investor sentiment index and the partial least squares sentiment index. Asset allocation analysis indicates the forecasting power of TSENT is economically valuable to investors. Finally, we show that the time-series bond risk premia predictability associated with TSENT relates to its predictive power for macro-economic performance, such as payroll employment, unemployment rate, and industrial production.

JEL CLASSIFICATION

G11, G12, G17

1 | INTRODUCTION

Treasury bonds play a central role in investors' asset allocation. Their risk and return dynamics have received considerable attention from financial researchers. An extensive literature demonstrates the predictability of Treasury bonds. For instance, predictors based on forward spreads (Fama & Bliss, 1987), yield spreads (Campbell & Shiller, 1991), a linear combination of forward rates (Cochrane & Piazzesi, 2005), and macro factors (Ludvigson &

Ng, 2009) possess significant in-sample predictive power for Treasury bond excess returns with maturities of 2–5 years. Gargano et al. (2019) find both statistically and economically significant Treasury bond return predictability after accounting for estimation error and model uncertainty. In general, ample studies highlight the role of financial and macro factors in predicting bond premia.

Behavioral predictors, such as marketwide investor sentiment, also hold predictive power for aggregate assets returns (e.g., Baker & Wurgler, 2007; Baker et al., 2012; Doukas & Milonas, 2004; Huang et al., 2015). Recent research focuses mainly on investor sentiment effects in the stock market and posits that investor sentiment affects financial markets by driving asset prices away from fundamentals (De Long et al., 1990; Shefrin & Statman, 1994). The subsequent price reversals lead to a negative relation between investor sentiment and future returns (Brown & Cliff, 2004; Garcia, 2013; Jiang et al., 2019; Schmeling, 2009). However, several recent studies document that marketwide investor sentiment not only contains irrational factors but also includes information about fundamentals and possesses some positive predictive power about stock returns (e.g., Bollen et al., 2011; Gao et al., 2021).

Although these two concepts are prominent in their academic fields, there is little research that explores the role of investor sentiment in predicting bond premia in a unified framework. Consequently, several research questions remain unanswered: Can the investor sentiment that pertains to the Treasury market forecast the future excess returns of government bonds? Is the forecasting power of Treasury sentiment beyond that contained in conventional sovereign bond return predictors? Are forecasting gains from the Treasury sentiment economically valuable to investors? What drives the predictive power of Treasury sentiment?

We contribute to the literature by addressing all of these questions. Using the Treasury market sentiment (TSENT) of institutional investors provided by SENTIX (a leading behavioral finance research institution in market sentiment data), we find that TSENT positively predicts 1-month-ahead Treasury bond excess returns in sample. Specifically, a 1 SD increase in TSENT leads to 9.1, 12.2, 14.3, and 15.6 basis point (bps) increases for 2-, 3-, 4-, and 5-year Treasury returns in the subsequent month, respectively. We also find that TSENT's explanatory power decreases with bond maturity. The univariate predictive model R^2 decreases from 4.721% to 1.531% as bond maturity increases from 2 to 5 years. The results are qualitatively similar when using excess bond returns based on the Gürkaynak et al. (2007) method or Fama Treasury bond portfolio returns from the Center for Research in Security Prices (CRSP) database. The positive relation between TSENT and future Treasury returns is robust after adopting bootstrapping exercises and the weighted least squares method of Yohai (1987).

The positive predictive power of TSENT indicates that TSENT contains valuable information about the future price movement of Treasury bonds, which has not been incorporated into current prices. This finding is different from that documented in most of the stock market sentiment literature, which argues that investor sentiment is irrational, and its negative impact on future stock index returns is due to stock price reversal after high sentiment. The contradicting implications may arise for two reasons. First, there exists segmentation between the pricing of debt and equity (Geanakoplos, 2010; Simsek, 2013) and sentiment may have distinct implications. For instance, López-Salido et al. (2017) point out that credit market sentiment is fundamentally different from stock market sentiment. They show that only high credit market sentiment forecasts macroeconomic slowdown. Therefore, unlike stock market sentiment that reflects irrational behaviors among investors, it is possible that TSENT contains fundamental-related information. Second, we obtain our Treasury sentiment measure from sophisticated institutional investors, whereas most equity studies use sentiment measures from retail investors. It is well recognized that institutional investors have an information advantage (Gao et al., 2021; Hendershott et al., 2015) because they are more likely to have a cohesive level of professional knowledge, rich resources at their disposal, and an abundant amount of time available to perform intensive analysis.

However, the forecasting power of Treasury sentiment could be due to the incorporation of other relevant information known to predict Treasury bond excess returns. To address this concern, we compare our Treasury sentiment with a list of conventional bond return predictors, such as Fama and Bliss (1987) forward spreads (FB),

Cochrane and Piazzesi (2005) forward rate factor (CP), and Ludvigson and Ng (2009) macro factor (LN). We show that this is not the case. In fact, the coefficient on TSENT is positive and significant after controlling for FB, CP, and LN. Meanwhile, only the predictive power of LN survives in the kitchen-sink type of regression.

Welch and Goyal (2008) show that most predictors that have strong in-sample predictive power fail to predict the equity risk premium out of sample. Therefore, we test the out-of-sample performance of TSENT. First, following Welch and Goyal (2008), we compare the out-of-sample forecasting accuracy of a predictive model with our Treasury sentiment to that of a common benchmark forecasting model. The results show that Treasury sentiment consistently provides substantial out-of-sample forecasting gains relative to a prevailing mean benchmark or a first-order autoregressive model. Second, forecast-encompassing tests soundly support that the TSENT-based forecast contains information that is not reflected in the CP, LN, and FB-based forecasts. All of these findings indicate that Treasury sentiment is important in predicting bond premia.

Meanwhile, some studies focus on the economic value of Treasury bond return predictability and find a puzzling disparity. Although FB and CP are powerful predictors, Thornton and Valente (2012) find they do not generate any significant economic value for investors. Exploiting a sophisticated estimation strategy for affine term structure models that jointly fits yields and bond excess returns, Sarno et al. (2016) document that the high forecast accuracy associated with the model cannot outperform the expectations hypothesis out of sample in terms of economic value. Gargano et al. (2019) argue that part of the puzzle is the use of a simple predictive regression model. Hence, it is essential to explore whether Treasury sentiment is an economically meaningful predictor because a return predictor with low economic value may not be attractive to investors.

In this article, we examine the economic significance of TSENT's predictive power from an asset allocation perspective and find that TSENT-based forecasts provide the highest certainty equivalent return (CER) gains that range from 42 to 264 bps. The Sharpe ratios associated with TSENT-based forecasts are also attractive and higher than those of CP-, LN-, and FB-based forecasts.

In a further investigation, we assess possible sources of the return predictability with Treasury sentiment. We examine the informational role of Treasury sentiment by looking at three important economic indicators: payroll employment, civilian unemployment rate, and industrial production. We find that the optimism in Treasury markets can successfully predict decreases in payroll employment change and industrial production, and increases in unemployment rate growth. Results suggest that TSENT contains valuable information for future macroeconomic performance.

To our knowledge, studies closely related to ours are Laborda and Olmo (2014), Çepni et al. (2020), and Li (2021). Laborda and Olmo (2014) conduct perhaps the first major study to directly test whether subjective investors' expectations affect government bond returns. They find that equity market investor sentiment, proxied by the investor sentiment index of Baker and Wurgler (2006), has predictive power for Treasury returns. Accordingly, Çepni et al. (2020) show that an aligned investor sentiment index proposed by Huang et al. (2015) can predict the excess returns of US government bonds. Li (2021) explores the sentiment effects in emerging markets. Different from these stock market sentiment-based predictability studies, we directly use Treasury sentiment to explore its effect on Treasury bond returns and find that the role Treasury sentiment plays in forecasting government bond returns is distinct from that of stock market sentiment. Our findings are valuable to practitioners as well. Portfolio managers can leverage our results to have more predictable government bond returns in their portfolios. Thus, in portfolio management, the uncertainty related to the returns of different asset compositions is significantly lower, leading to better dynamics of asset allocation.

2 | DATA

In this section, we introduce the Treasury sentiment proxy, stock market sentiment proxy, Treasury returns, and three conventional bond return predictors. Additionally, we provide a data statistics summary.

2.1 | Treasury sentiment measure

We adopt the 10-year US Treasury sentiment index from SENTIX.¹ Every week, SENTIX surveys approximately 200–300 institutional investors from a pool of more than 4000 SENTIX members² about their expectations on markets and economic trends over the next 6 months (referred to as the strategic bias of institutional investors by SENTIX). Among institutional investors are traders, analysts, fund managers, and economists, who register the survey with a verified official e-mail address linking to recognized financial institutions, such as a commercial/investment bank, an insurance company, or a mutual/hedge firm. Although these representatives of institutions volunteer to provide their perceptions about future asset prices, SENTIX provides effective incentives to encourage its registers to be frequent survey participants.

Since its inception in February 2001, SENTIX has been providing information about investors' estimates of 10-year US government bond price trends. Hence, our sample period spans February 2001 to December 2019. Similar to other survey-based sentiment data (e.g., American Association of Individual Investors [AAII] sentiment measure), our Treasury sentiment proxy, the SENTIX strategic bias index (TSENT hereafter) of US bonds, is constructed as the difference between the percentage of all respondents who have bullish views for future government bond returns and the percentage who have bearish views. Therefore, a positive (negative) sentiment suggests that investors in general are optimistic (pessimistic) about the future performance of the Treasury market. Although SENTIX data are published weekly, most studies focus on the predictability of Treasury bond excess returns on a monthly basis. Therefore, we convert the weekly sentiment index into a monthly index by taking the average of weekly data within a given month.

We also compare the performance of TSENT with the stock market sentiment index in predicting bond risk premia. Specifically, we include the investor sentiment index of Baker and Wurgler (2006, 2007) (BW hereafter) and the partial least squares (PLS) sentiment index of Huang et al. (2015) (HJTZ hereafter). BW is based on the first principal component of five stock market trading statistics and has been shown to explain the movements of the US equity market.³ Meanwhile, Huang et al. (2015) apply the PLS approach first to filter out the irrelevant noise component of the stock market trading statistics used in Baker and Wurgler (2006, 2007). Afterward, they perform principal component analysis, and the first principal component is HJTZ, which exhibits stronger predictive power than BW in and out of sample.⁴

2.2 | Treasury returns

We use the Treasury returns calculated from the parameters estimated by Gürkaynak et al. (2007) in our main analysis (see, e.g., Gargano et al., 2019). Specifically, we define the time t zero-coupon log yield on a bond maturing in n years, $y_t^{(n)}$, as

$$y_t^{(n)} = \beta_{0t} + \beta_{1t} \frac{1 - \exp\left(-\frac{n}{\tau_1}\right)}{\frac{n}{\tau_1}} + \beta_{2t} \left[\frac{1 - \exp\left(-\frac{n}{\tau_1}\right)}{\frac{n}{\tau_1}} - \exp\left(-\frac{n}{\tau_1}\right) \right] + \beta_{3t} \left[\frac{1 - \exp\left(-\frac{n}{\tau_2}\right)}{\frac{n}{\tau_2}} - \exp\left(-\frac{n}{\tau_2}\right) \right]. \quad (1)$$

¹SENTIX has 400 sentiment indices across asset categories and regions. The data are widely used in the literature (e.g., Gao et al., 2021; Heiden et al., 2013; Schmeling, 2007).

²Among all SENTIX members, 75% are individual investors and 25% are institutional investors.

³The BW index is available from Jeffrey Wurgler's website: <http://people.stern.nyu.edu/jwurgler/>. We use the orthogonalized BW index.

⁴The HJTZ index is available from Guofu Zhou's website: <http://apps.olin.wustl.edu/faculty/zhou/zpublications.html>. We use the orthogonalized HJTZ index.

The parameters $(\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$ are provided by Gürkaynak et al. (2007).⁵ Following the literature, we consider bond portfolios with 2-, 3-, 4-, and 5-year maturities and use the last day of each month's estimated log yields as bond-month yields.

The monthly raw returns and excess returns, relative to the 1-month Treasury bill rate, are, respectively:

$$ret_{t+1}^{(n)} = p_{t+1}^{(n-\frac{1}{12})} - p_t^{(n)} = ny_t^{(n)} - \left(n - \frac{1}{12}\right)y_{t+1}^{(n-\frac{1}{12})}, \quad (2)$$

$$exret_{t+1}^{(n)} = ret_{t+1}^{(n)} - \frac{1}{12}y_t^{(\frac{1}{12})}, \quad (3)$$

where $p_t^{(n)}$ is the logarithm of the time t price of a bond with n years to maturity. For robustness, we also use the constant-maturity Fama Treasury bond portfolio returns from the CRSP database.⁶

2.3 | Conventional Treasury return predictor variables

The conventional predictors consist of the Fama–Bliss forward spreads (FB), Cochrane and Piazzesi forward rate factor (CP), and Ludvigson and Ng (2009) macro factor (LN). Specifically, FB is given by

$$FB_t^{(n)} = f_t^{(n-\frac{1}{12}, n)} - \frac{1}{12}y_t^{(\frac{1}{12})} = ny_t^{(n)} - \left(n - \frac{1}{12}\right)y_t^{(n-\frac{1}{12})} - \frac{1}{12}y_t^{(\frac{1}{12})}, \quad n \in (2, 3, 4, 5), \quad (4)$$

where f_t is the forward rates. We compute FB for 2-, 3-, 4-, and 5-year maturities government bonds.

Cochrane and Piazzesi (2005) estimate a predictive regression where bond excess returns are regressed on the full set of forward rates. Specifically, the predictive regression is

$$\frac{1}{4} \sum_{n=2}^5 exret_{t+1}^{(n)} = \gamma_0 + \gamma_1 f_t^{(1-\frac{1}{12}, 1)} + \gamma_2 f_t^{(2-\frac{1}{12}, 2)} + \gamma_3 f_t^{(3-\frac{1}{12}, 3)} + \gamma_4 f_t^{(4-\frac{1}{12}, 4)} + \gamma_5 f_t^{(5-\frac{1}{12}, 5)} + \varepsilon_{t+1}, \quad (5)$$

and the CP factor is given by the fitted value. Cochrane and Piazzesi (2005) find that this single factor, which is a linear combination of forward rates, has high predictive power on excess returns of Treasury bonds.

Ludvigson and Ng (2009) use macro factors to predict bond excess returns, and the macro factors are extracted from 132 monthly economic series via principal component analysis. Following Ludvigson and Ng (2009), the LN factor is the fitted value of the following predictive regression

$$\frac{1}{4} \sum_{n=2}^5 exret_{t+1}^{(n)} = \delta_0 + \delta_1 F_{1t} + \delta_2 F_{1t}^3 + \delta_3 F_{3t} + \delta_4 F_{4t} + \delta_5 F_{8t} + \varepsilon_{t+1}, \quad (6)$$

where $[F_{1t}, F_{3t}, F_{4t}, F_{8t}]$ is a subset of the first eight principal components of Ludvigson and Ng (2009).⁷

⁵The data are available at <https://www.federalreserve.gov/pubs/feds/2006/200628/200628abs.html>.

⁶The unique identifiers for the bond portfolios are 2000041, 2000042, 2000043, 2000044 for 2-, 3-, 4-, and 5-year maturities, respectively. The unique identifier for the 1-month risk-free rate is 2000001.

⁷The principal components data are downloaded from <https://www.sydneyludvigson.com/data-and-appendixes>.

2.4 | Data summary

Panel A of Table 1 reports summary statistics for the four monthly excess return series that calculated from the Treasury yield data of Gürkaynak et al. (2007). The mean and standard deviation of monthly excess returns increase with maturity. For instance, the 2-year bond returns have a mean of 0.087% and standard deviation of

TABLE 1 Summary statistics.

Panel A: 1-month excess returns (Gürkaynak et al. (2007), method)							
	2 years		3 years		4 years		5 years
Mean	0.087		0.146		0.200		0.248
SD	0.435		0.729		1.021		1.308
Max.	1.651		2.406		3.481		4.900
Min.	-1.333		-2.319		-3.171		-4.058
Panel B: 1-month excess returns (Fama bond portfolio data)							
	2 years		3 years		4 years		5 years
Mean	0.078		0.132		0.183		0.221
SD	0.307		0.569		0.835		1.100
Max.	1.424		2.109		2.768		4.279
Min.	-0.826		-1.718		-2.547		-3.310
Panel C: Predictors							
	FB2	FB3	FB4	FB5	CP	LN	TSENT
Mean	0.061	0.104	0.142	0.174	0.173	0.172	-0.283
SD	0.067	0.091	0.108	0.122	0.214	0.223	0.158
Max.	0.292	0.341	0.370	0.402	1.136	1.268	0.235
Min.	-0.090	-0.088	-0.082	-0.072	-0.460	-0.997	-0.575
Panel D: Correlation matrix							
	FB2	FB3	FB4	FB5	CP	LN	TSENT
FB2	1.000						
FB3	0.963	1.000					
FB4	0.902	0.984	1.000				
FB5	0.848	0.955	0.992	1.000			
CP	0.447	0.528	0.552	0.549	1.000		
LN	-0.077	0.009	0.072	0.113	0.201	1.000	
TSENT	-0.348	-0.362	-0.355	-0.340	0.144	0.155	1.000

Note: This table reports the summary statistics of monthly bond excess returns for 2 to 5 years and predictors used in our study. In Panel A, returns are constructed using daily Treasury yield data from Gürkaynak et al. (2007). In Panel B, returns are taken from Fama bond portfolio data. Panel C reports summary statistics of the predictors: Fama–Bliss forward spreads (2, 3, 4, and 5 years) (FB), Cochrane–Piazzesi (2005) forward rate factor (CP), Ludvigson–Ng (2011) factor (LN), and SENTIX Treasury market sentiment (TSENT). Panel D reports the correlation matrix of the predictors. The sample period is February 2001 to December 2019 (BW and HJTZ are available only up to December 2018).

0.435%. The mean and standard deviation of the 5-year bond returns increase to 0.248% and 1.308%, respectively. Panel B presents summary statistics for Treasury excess returns of Fama bond portfolios, and we observe similar patterns.

Panel C of Table 1 provides summary statistics for TSENT along with FB, CP, and LN. TSENT has an average value of -0.283 , indicating surveyed institutional investors are generally pessimistic about the prospects of government bond prices over our sample period. Traditional bond return predictors all have positive means. The values of standard deviation and the max-min ranges suggest that all variables display great variation over time.

Panel D of Table 1 shows FBs are positively correlated with CP, with correlations ranging between 0.447 and 0.552. However, they have a low correlation with LN, with correlations ranging between -0.077 and 0.113. LN and CP have a positive correlation. Meanwhile, TSENT demonstrates a negative association with FBs and is weakly correlated with CP and LN. These results indicate that TSENT may have distinct impacts on the Treasury bond market separate from conventional bond return predictors.

3 | PREDICTIVE REGRESSION ANALYSIS

3.1 | In-sample tests

In this subsection, we assess the predictive performance of TSENT on next-month Treasury excess returns using the following model:

$$Ret_{t+1}^{(n)} = a_0 + a_1 TSENT_t + a_2 CP_t + a_3 LN_t + a_4 FB_t^{(n)} + e_{t+1}, \quad n \in (2, 3, 4, 5), \quad (7)$$

where $Ret_{t+1}^{(n)}$ is 1-month-ahead excess returns for a Treasury bond portfolio with n years to maturity. $TSENT_t$ is Treasury market sentiment from SENTIX. It is our key variable of interest. The model also controls three conventional bond return predictors: CP, LN, and FB. All independent variables are standardized by their sample means and standard deviations. We run the model using the Newey–West (1987) heteroskedastic and autocorrelation robust covariance matrix. Table 2 presents the in-sample predictive regression results.

In Panel A of Table 2, the dependent variable is 1-month-ahead Treasury bond excess returns constructed using daily Treasury yield data from Gürkaynak et al. (2007). We find that Treasury sentiment consistently exerts strong positive forecasting power over next-month Treasury bond excess returns with all maturities with/without conventional bond return predictors. The predictability is both statistically and economically sizable. In the univariate setting, the values of the coefficient estimate on TSENT indicate that a 1SD increase in bond sentiment leads to 9.1, 12.2, 14.3, and 15.6 bps increases for 2-, 3-, 4-, and 5-year Treasury bond returns in the subsequent month, respectively. Moreover, after controlling for traditional return predictors, the forecasting power of TSENT persists at a conventional level. Thus, TSENT provides substantial information that is useful for forecasting Treasury bond excess returns beyond the information already contained in other well-known predictors. By itself, LN demonstrates significant predictive power for future Treasury returns in the predictive model with TSENT. FB and CP have no noticeable effects on next-month bond returns.⁸

Because monthly returns inherently contain a large unpredictable component, the R^2 statistics from the monthly in-sample predictive model are necessarily small. Nevertheless, the R^2 values show that TSENT alone explains 1.531%–4.721% of the variation in bond returns. After including CP, LN, and FB, all return predictors

⁸Consistent with the literature, the three conventional bond return predictors all display significant forecasting power over next-month Treasury returns with univariate regressions. The results are available from the authors upon request.

TABLE 2 In-sample forecasts.

	2 years		3 years		4 years		5 years	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A: 1-month excess returns (Gürkaynak et al. (2007), method)</i>								
TSENT	0.091** (0.036)	0.090*** (0.028)	0.122** (0.053)	0.127*** (0.048)	0.143** (0.069)	0.135** (0.067)	0.156* (0.083)	0.130 (0.085)
CP		0.056 (0.045)		0.084 (0.075)		0.127 (0.105)		0.180 (0.131)
LN		0.101*** (0.029)		0.151*** (0.054)		0.201** (0.081)		0.253** (0.111)
FB		0.061 (0.040)		0.102 (0.068)		0.116 (0.090)		0.113 (0.110)
R ² (%)	4.721	16.217	2.996	13.284	2.105	11.685	1.531	10.704
<i>Panel B: 1-month excess returns (Fama bond portfolio data)</i>								
TSENT	0.073*** (0.027)	0.067*** (0.019)	0.110** (0.044)	0.111*** (0.037)	0.128** (0.057)	0.123** (0.053)	0.145** (0.072)	0.122* (0.070)
CP		0.044 (0.033)		0.064 (0.060)		0.097 (0.086)		0.155 (0.114)
LN		0.076*** (0.021)		0.117*** (0.041)		0.163** (0.064)		0.214** (0.093)
FB		0.035 (0.030)		0.079 (0.056)		0.097 (0.075)		0.094 (0.094)
R ² (%)	5.988	17.972	3.977	13.943	2.498	11.651	1.856	11.074

Note: This table reports the in-sample predictive regression results. The predictors include SENTIX Treasury market sentiment (TSENT), Fama–Bliss forward spreads (2, 3, 4, and 5 years) (FB), Cochrane–Piazzesi (2005) forward rate factor (CP), and Ludvigson–Ng (2011) factor (LN). In Panel A, the dependent variables are 1-month excess returns constructed using daily Treasury yield data from Gürkaynak et al. (2007). In Panel B, the dependent variables are 1-month excess returns taken from Fama bond portfolio data. All predictors are standardized using their respective sample means and sample standard deviations. The regressions are estimated using the Newey–West (1987) heteroskedastic and autocorrelation robust covariance matrix. Standard errors are provided in parentheses. The sample period is February 2001 to December 2019.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

together explain 10.704%–16.217% of total variation in next-month Treasury returns. Moreover, results show that TSENT's explanatory power decreases with bond maturity in the univariate regressions. Quantitatively, R^2 monotonically decreases from 4.721% to 1.531%. Therefore, the predictive power is weaker for long-term Treasury bonds.

In Panel B of Table 2, we adopt 1-month excess returns taken from Fama bond portfolio data as a robustness check. It is clear that using alternative bond excess returns as the dependent variables does not qualitatively affect our findings in Panel A. Hence, Treasury sentiment is a strong and robust predictor for future Treasury excess returns.

3.2 | Out-of-sample tests

3.2.1 | Out-of-sample R^2

Welch and Goyal (2008) argue that the superior in-sample performance of a return predictor cannot guarantee its notable performance out of sample. They show that most aggregate stock return predictors that survive from in-sample tests cannot beat the historical average benchmark out of sample. Therefore, we conduct standard out-of-sample predictability tests to examine whether our proxy for Treasury sentiment generates incremental forecasting power over next-month Treasury returns relative to a benchmark model.

In light of Welch and Goyal (2008), we use three steps to conduct out-of-sample predictability tests.⁹ First, we employ an expanding estimation window to compute out-of-sample forecasts of the Treasury return:

$$\widehat{Ret}_{t+1}^m = \hat{a}_{0,m} + \hat{a}_{1,m} TSENT_{t+1}, \quad (8)$$

where $\hat{a}_{0,m}$ and $\hat{a}_{1,m}$ are the coefficient estimates obtained through regressing excess Treasury returns on lagged Treasury sentiment with an intercept for an in-sample period with m observations in the estimation window. For robustness, we choose the first 3, 5, and 10 years as our initial in-sample estimating periods.

Second, we calculate the out-of-sample R^2 statistics (R_{OS}^2), which measures the proportional reduction in the mean square prediction error (MSPE) of our predictive regression, relative to a benchmark forecast.

$$R_{OS}^2 = 1 - \frac{\sum_{k=q_0+1}^q (Ret_{m+k} - \widehat{Ret}_{m+k})^2}{\sum_{k=q_0+1}^q (Ret_{m+k} - \bar{Ret}_{m+k})^2}, \quad (9)$$

where $\widehat{Ret}_{m+k}$ denotes the expected value of Treasury returns before period $m+k$ from a baseline model. We select two common benchmark models: (1) prevailing mean benchmark model ($Ret_{m+1} = \alpha + \varepsilon_{t+1}$) and (2) first-order autoregressive model ($Ret_{m+1} = \alpha + \beta Ret_m + \varepsilon_{t+1}$). q_0 and q represent the number of observations in our in-sample period and the overall sample period, respectively. A positive R_{OS}^2 indicates our investigated predictor generates lower forecasting errors than the benchmark model.

Finally, we use the Clark–West (2007) out-of-sample MSPE-adjusted statistic to evaluate R_{OS}^2 . Specifically, a large statistic suggests rejection of the null hypothesis that $R_{OS}^2 \leq 0$ (alternative hypothesis: $R_{OS}^2 > 0$).¹⁰ The results in Table 3 show that all R_{OS}^2 values are positive and significant, with magnitudes that ranging from 1.55% to 8.42%. There is strong evidence that Treasury sentiment consistently produces out-of-sample forecasting gains relative to the baseline model. The sizes of R_{OS}^2 are also economically significant, as Campbell and Thompson (2008) state that a monthly out-of-sample R^2 of approximately 0.5% is already big enough to generate significant economic value.

3.2.2 | Forecast encompassing tests

In this section, we conduct forecast encompassing tests to provide a formal evaluation for the performance of different predictors. Specifically, we first form an optimal combination forecast as a convex combination of a predictive regression forecast based on one of the three conventional predictors and the predictive regression forecast based on TSENT:

$$\hat{r}_{t+1}^* = (1 - \rho) \hat{r}_{t+1}^i + \rho \hat{r}_{t+1}^{TSENT}, \quad (10)$$

⁹The out-of-sample tests are widely used in the return predictability literature (e.g., Huang et al., 2015; Jiang et al., 2019; Rapach et al., 2010).

¹⁰The test uses one-tail standard normal critical values (1.282 for 10%, 1.645 for 5%, 2.327 for 1%) for the Clark–West (2007) MSPE-adjusted statistics.

TABLE 3 Out-of-sample forecasts.

	Gürkaynak et al. (2007)				Fama bond portfolio data			
	PM		AR(1)		PM		AR(1)	
	R_{Os}^2	CW	R_{Os}^2	CW	R_{Os}^2	CW	R_{Os}^2	CW
<i>Panel A: 2 years</i>								
3 years	5.33***	3.88	5.01***	3.25	6.83***	4.53	6.44***	3.77
5 years	3.75***	3.14	2.63***	2.43	4.90***	3.70	2.87***	2.94
10 years	8.42***	3.11	11.51***	3.74	23.53***	5.17	21.49***	5.05
<i>Panel B: 3 years</i>								
3 years	3.45***	2.99	3.64***	2.73	4.68***	3.54	3.99***	2.86
5 years	2.84***	2.48	2.75**	2.15	3.80***	2.95	2.48**	2.16
10 years	3.77**	2.09	5.93***	2.64	11.43***	3.38	11.87***	3.53
<i>Panel C: 4 years</i>								
3 years	2.32***	2.40	2.74***	2.39	2.78***	2.64	3.30***	2.61
5 years	2.12**	2.07	2.43**	2.00	2.55**	2.28	2.72**	2.16
10 years	2.09*	1.58	3.22**	1.95	5.01**	2.20	6.26***	2.52
<i>Panel D: 5 years</i>								
3 years	1.55**	1.95	1.98**	2.05	1.55**	1.95	1.98**	2.05
5 years	1.54**	1.76	2.05**	1.90	1.54**	1.76	2.05**	1.90
10 years	2.51*	1.60	1.84*	1.50	2.51*	1.60	1.84*	1.50

Note: This table reports the proportional reduction in mean squared forecast error (MSFE) at the 1-month horizon for a predictive regression forecast of Treasury monthly excess returns (2–5 years) based on SENTIX Treasury market sentiment (TSENT) vis-à-vis the prevailing mean (PM) benchmark forecast and AR(1) model, where statistical significance is based on the Clark–West (2007) statistic for testing the null hypothesis that the prevailing mean MSFE or AR(1) MSFE is greater than the predictive regression MSFE. The in-sample periods are 3 years, 5 years, and 10 years, respectively.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

where $\hat{r}_{t+1}^i$ is the predictive regression forecast based on CP, LN, or FB, and $\hat{r}_{t+1}^{\text{TSENT}}$ is the predictive regression forecast based on TSENT. The weight ρ is restricted between 0 and 1. If $\rho = 1$, the optimal combination forecast relies solely on the forecast based on TSENT, and the predictive regression forecast based on TSENT encompasses the predictive regression forecast based on other three conventional predictors. Alternatively, if $\rho = 0$, TSENT does not contain information that is useful for forecasting Treasury bond excess returns beyond the information already found in CP, LN, and FB. To test the significance of ρ , we calculate the Harvey–Leybourne–Newbold (HLN) statistic (Harvey et al., 1998).

Table 4 reports the results of encompassing tests based on monthly return forecasts for Treasury bonds with different maturities. Results in Panels A–D show that for bond excess returns for 2- to 5-year maturities, though less than one, the weights on TSENT-based forecast are all sizable and significant (8 of 12 have values above 0.6). Hence, the information content of TSENT is not subsumed by conventional predictors. For Treasury bonds with Fama bond return series, most ρ estimates are sizable and significant (larger than 0.6 in 10 of 12 estimates). In sum, we find strong evidence that TSENT contains superior information content relative to the other three conventional predictors (CP, LN, and FB) used in the literature.

TABLE 4 Forecast encompassing tests.

	Gürkaynak et al. (2007)		Fama bond portfolio data	
	ρ	HLN	ρ	HLN
<i>Panel A: 2 years</i>				
CP	0.64***	3.35	0.62***	3.54
LN	0.68***	2.76	0.69***	3.29
FB	0.70***	3.27	0.72***	3.55
<i>Panel B: 3 years</i>				
CP	0.62***	2.76	0.64***	3.09
LN	0.71**	2.38	0.74***	2.66
FB	0.66***	2.88	0.68***	3.13
<i>Panel C: 4 years</i>				
CP	0.55**	2.19	0.61**	2.45
LN	0.75**	2.17	0.79**	2.30
FB	0.59**	2.39	0.63***	2.63
<i>Panel D: 5 years</i>				
CP	0.45	1.61	0.48*	1.81
LN	0.81*	1.93	0.74**	2.01
FB	0.51*	1.89	0.55**	2.07

Note: This table reports the estimated weight on the predictive regression forecast of Treasury monthly excess returns (2–5 years) based on SENTIX Treasury market sentiment (TSENT) in a combination forecast that takes the form of a convex combination of a predictive regression forecast based on TSENT and a predictive regression forecast based on Fama–Bliss forward spreads (2, 3, 4, and 5 years) (FB), Cochrane–Piazzesi (2005) forward rate factor (CP), and Ludvigson–Ng (2011) factor (LN). The statistical significance is based on the Harvey–Leybourne–Newbold (HLN) statistic (Harvey et al., 1998) for testing the null hypothesis that the weight (ρ) on the TSENT-based forecast is equal to zero against the alternative hypothesis that the weight on the TSENT-based forecast is greater than zero.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

3.3 | Robustness checks

To further validate our empirical results, we perform a battery of robustness checks. We first compare the predictive power of TSENT with equity market sentiment proxies such as the BW index and HJTZ index. Next, we confirm that our main findings are not due to small-sample issues.

In Table 5, we compare the predictive power between TSENT and equity market sentiment. Specifically, we modify Equation (7) by incorporating BW and HJTZ separately. Results show that the predictive power associated with TSENT is beyond that of BW and HJTZ. Meanwhile, BW and HJTZ fail to predict the 1-month-ahead Treasury bond excess returns.¹¹ Results are consistent with the view that a segmentation exists between credit market sentiment and stock market sentiment (López-Salido et al. 2017).

¹¹Similar to Laborda and Olmo (2014) and Çepni et al. (2020), we find that equity sentiment measures positively predict next-month Treasury returns using univariate analysis.

TABLE 5 Treasury sentiment vis-à-vis equity sentiment.

	2 years		3 years		4 years		5 years	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A: 1-month excess returns (Gürkaynak et al. (2007), method)</i>								
TSENT	0.091*** (0.028)	0.084*** (0.030)	0.135*** (0.048)	0.125** (0.053)	0.158** (0.068)	0.147* (0.077)	0.166* (0.087)	0.153 (0.098)
CP	0.056 (0.045)	0.050 (0.050)	0.086 (0.074)	0.086 (0.080)	0.134 (0.102)	0.136 (0.108)	0.194 (0.128)	0.199 (0.134)
LN	0.101*** (0.029)	0.098*** (0.031)	0.151*** (0.054)	0.152*** (0.056)	0.200** (0.081)	0.206** (0.085)	0.253** (0.112)	0.262** (0.115)
FB	0.061 (0.041)	0.058 (0.040)	0.099 (0.069)	0.103 (0.070)	0.106 (0.091)	0.123 (0.096)	0.094 (0.112)	0.128 (0.120)
BW	-0.003 (0.032)		-0.025 (0.058)		-0.051 (0.083)		-0.082 (0.108)	
HJTZ		0.015 (0.039)		-0.006 (0.068)		-0.029 (0.097)		-0.056 (0.127)
R ² (%)	16.219	16.271	13.359	13.287	11.836	11.721	10.936	10.790
<i>Panel B: 1-month excess returns (Fama bond portfolio data)</i>								
TSENT	0.066*** (0.019)	0.060*** (0.021)	0.119*** (0.037)	0.108*** (0.040)	0.138** (0.055)	0.127** (0.060)	0.151** (0.074)	0.131 (0.081)
CP	0.044 (0.033)	0.037 (0.037)	0.065 (0.059)	0.062 (0.064)	0.102 (0.084)	0.100 (0.089)	0.166 (0.111)	0.163 (0.116)
LN	0.076*** (0.021)	0.073*** (0.022)	0.117*** (0.041)	0.116*** (0.043)	0.162** (0.065)	0.164** (0.067)	0.214** (0.093)	0.217** (0.097)
FB	0.035 (0.030)	0.031 (0.029)	0.077 (0.057)	0.077 (0.056)	0.090 (0.077)	0.099 (0.078)	0.078 (0.095)	0.099 (0.102)
BW	0.003 (0.023)		-0.017 (0.044)		-0.032 (0.066)		-0.065 (0.091)	
HJTZ		0.017 (0.028)		0.006 (0.052)		-0.010 (0.078)		-0.022 (0.107)
R ² (%)	17.976	18.123	13.996	13.948	11.742	11.657	11.281	11.092

Note: This table reports the in-sample predictive regression results. The predictors include SENTIX Treasury market sentiment (TSENT), Fama–Bliss forward spreads (2, 3, 4, and 5 years) (FB), Cochrane–Piazzesi (2005) forward rate factor (CP), Ludvigson–Ng (2011) factor (LN), Baker–Wurgler (2006, 2007) investor sentiment index (BW), and Huang et al. (2015) partial least squares sentiment index (HJTZ). The forecast horizon is monthly. In Panel A, the dependent variables are 1-month excess returns constructed using daily Treasury yield data from Gürkaynak et al. (2007). In Panel B, the dependent variables are 1-month excess returns taken from Fama bond portfolio data. The sample period is February 2001 to December 2018. All predictors are standardized using their respective sample means and sample standard deviations. The regressions are estimated using the Newey–West (1987) heteroskedastic and autocorrelation robust covariance matrix. Standard errors are provided in parentheses.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

TABLE 6 Robustness tests for small sample bias.

	Gürkaynak et al. (2007)		Fama bond portfolio data	
	β	SE	β	SE
<i>Panel A: Bootstrapped</i>				
2 years	0.097***	0.027	0.077***	0.019
3 years	0.131***	0.045	0.119***	0.034
4 years	0.153**	0.063	0.138***	0.049
5 years	0.175**	0.072	0.160**	0.064
<i>Panel B: MM test</i>				
2 years	0.072**	0.028	0.055***	0.017
3 years	0.114**	0.047	0.123***	0.034
4 years	0.140**	0.066	0.134**	0.054
5 years	0.139	0.087	0.129*	0.074

Note: Panel A reports means and standard deviations of the estimated slope coefficients on SENTIX Treasury market sentiment (TSENT) in Model (7) across bootstrap replications. We randomly draw with replacement 1000 samples of Treasury returns and return predictors. The size of the random samples is equal to the number of full-sample observations. We then estimate the regression in Equation (7) for each random sample and examine the empirical distribution of the coefficient estimates. Panel B reports the estimated slope coefficients on TSENT in Model (7) using the MM weighted least squares method of Yohai (1987). This procedure is robust in the presence of outliers. The sample period is February 2001 to December 2019.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

To address potential small-sample bias, we conduct two tests. First, we follow Lucca and Moench (2015) to conduct a simple bootstrapping exercise. Specifically, we generate 1000 samples with replacement of Treasury returns, Treasury sentiment, and conventional Treasury return predictors. Each random sample has the same size as the sample used in the baseline regression of Equation (7). We then run Equation (7) and report the empirical distribution of the coefficient estimates on Treasury sentiment. The results in Panel A of Table 6 show that the mean and standard deviation of the estimated coefficients on Treasury sentiment are close to the corresponding estimates reported in Table 2.

Second, we use the MM weighted least squares method of Yohai (1987) to replace the traditional ordinary least squares method when estimating Equation (7). The MM estimator is robust to a sample with outliers. Panel B of Table 6 shows that our results are not driven by extreme observations.

4 | ASSET ALLOCATION AND ECONOMIC VALUE

A limitation to the $R_{O_S}^2$ metric is that it does not explicitly account for the risk borne by an investor over the out-of-sample period. To address this, following Campbell and Thompson (2008) and Rapach et al. (2010), we measure the economic value of TSENT's predictive ability from an asset allocation perspective. Specifically, we measure the economic value by the realized utility gain for a mean–variance investor on a real-time basis, who allocates between Treasury bonds and risk-free bills. The optimal weight on the n -year Treasury bond is given as follows:

$$w_t^{(n)} = \frac{1}{Y} \frac{\hat{r}_{t+1}}{\hat{\sigma}_{t+1}^2}, \quad (11)$$

where γ is the risk aversion coefficient of the investor, $\hat{r}_{t+1}$ is the forecast n -year Treasury excess return from the predictive regression, and $\hat{\sigma}_{t+1}^2$ is a forecast of the excess return variance, which is given by the volatility forecast using a 4-year moving window of past returns.¹²

The realized utility, or CER, of the investor over the out-of-sample period is

$$\text{CER}^{(n)} = \bar{R} - 0.5\gamma\sigma^2, \quad (12)$$

where $\bar{R}$ and σ^2 are the mean and variance, respectively, of the return of the portfolio that allocates $w_t^{(n)}$ to the n -year Treasury bond and $(1 - w_t^{(n)})$ to the risk-free bills over the forecast evaluation period. Then we measure the utility gain as the difference between the CER for an investor when the predictive regression forecast is used to make investment decisions and the CER when the prevailing mean benchmark forecast is used. The utility gain can be interpreted as the fee an investor is willing to pay to obtain the predictive regression forecast in place of the prevailing mean forecast. In our empirical analysis, we annualize the CER gain and focus on the monthly forecast horizon.

The first column of Table 7 presents the CER gains from a forecast using the predictive regression over using the historical mean for the 2006:01 to 2019:12 out-of-sample period. Panel A reports the results for the 2-year Treasury bond, where predictor TSENT generates a positive CER gain of 42 bps. In contrast, CP, LN, and FB generate smaller or even negative CER gains (−77, 22, and −162 bps, respectively). Moreover, the buy-and-hold portfolio that passively holds a 2-year Treasury bond produces an abysmal negative CER gain of −356 bps. The results for the 3-year Treasury bond are provided in Panel B. TSENT continues to deliver a positive CER gain of 93 bps, whereas other predictors still produce smaller or negative values. For the 5-year Treasury bond, FB generates a sizable CER gain of 200 bps; however, the CER gain associated with TSENT is even larger. Results suggest that investors are better off following an investment strategy based on the forecasts generated by TSENT.

The third column of Table 7 reports the CER gains where the Treasury returns data are from Fama bond portfolios. For Treasury bonds with maturities of 3–5 years, TSENT always provides the highest and positive CER gains, ranging from 39 bps (3-year Treasury bond) to 261 bps (5-year Treasury bond). However, the LN-based forecast provides higher CER gains for 2-year Treasury bond. Nevertheless, there is strong evidence that the economic value of TSENT's predictive ability is robust to alternate Treasury return series.

The second and fourth columns of Table 7 report annualized Sharpe ratios for the portfolios. The ratios for the portfolio based on the prevailing mean benchmark forecast decrease for Treasury bonds with longer maturities and range from 0.54 (5-year Treasury bond) to 0.69 (2-year Treasury bond), which are all below those of TSENT. Meanwhile, the Sharpe ratios of TSENT also beat the CP, LN, and FB for Treasury bonds with 2-, 3-, and 4-year maturities. For the 5-year Treasury bond, TSENT and FB generate the same Sharpe ratios of 0.66, but less than that of the buy-and-hold strategy by 0.03. When returns are replaced by the Fama bond portfolios, we observe even higher Sharpe ratios for TSENT. Moreover, across all bond maturities, the Sharpe ratios associated with TSENT are the highest among the individual predictors.

The results in this section indicate that there is substantial economic value in SENTIX. For a risk-averse investor that allocates between Treasury bonds and risk-free bills, making investment decisions based on SENTIX forecasts not only provides the highest CER gains but also yields sizable Sharpe ratios.

5 | TSENT AND BUSINESS CYCLES

Having documented that TSENT contains material information about bond risk premia, next we assess the potential sources of the predictive power associated with TSENT. Cochrane (2007) states that any economically compelling return predictor should be able to forecast future macroeconomic condition. Gilchrist and Zakrajšek (2012) and

¹²Following Thornton and Valente (2012) and Gargano et al. (2019), we assume a coefficient of risk aversion $\gamma = 5$. We also restrict the optimal weights $w_t^{(n)}$ to lie between −1.0 and 8.0.

TABLE 7 Out-of-sample CER gains and Sharpe ratios.

	Gürkaynak et al. (2007)		Fama bond portfolio data	
	CER gains	Sharpe ratios	CER gains	Sharpe ratios
<i>Panel A: 2 years</i>				
Prevailing mean		0.69		0.87
CP	-0.77	0.63	-0.50	0.83
LN	0.22	0.74	0.24	0.93
FB	-1.62	0.55	-1.23	0.81
TSENT	0.42	0.76	0.09	0.91
Buy and hold	-3.56	0.68	-4.27	0.87
<i>Panel B: 3 years</i>				
Prevailing mean		0.67		0.82
CP	-0.96	0.61	-0.94	0.76
LN	0.57	0.71	0.25	0.86
FB	-1.44	0.56	-1.51	0.72
TSENT	0.93	0.75	0.39	0.88
Buy and hold	-3.05	0.72	-5.11	0.84
<i>Panel C: 4 years</i>				
Prevailing mean		0.58		0.74
CP	-0.85	0.58	-0.96	0.69
LN	-0.36	0.61	0.07	0.75
FB	0.73	0.62	-0.92	0.68
TSENT	2.19	0.70	0.61	0.78
Buy and hold	-0.76	0.71	-3.56	0.81
<i>Panel D: 5 years</i>				
Prevailing mean		0.54		0.57
CP	-0.28	0.63	0.05	0.64
LN	-1.79	0.52	-1.01	0.58
FB	2.00	0.66	1.46	0.66
TSENT	2.64	0.66	2.61	0.70
Buy and hold	0.67	0.69	-0.10	0.73

Note: This table reports the annualized certainty equivalent return (CER) gain (in percent) for a mean–variance investor with relative risk aversion coefficient of 5 who allocates between Treasury bonds with maturities of 2 to 5 years and risk-free bills using a predictive regression excess return forecast based on the predictor variable in the first column relative to the prevailing mean benchmark forecast. The predictors include SENTIX Treasury market sentiment (TSENT), Fama–Bliss forward spreads (2, 3, 4, and 5 years) (FB), Cochrane–Piazzesi (2005) forward rate factor (CP), and Ludvigson–Ng (2011) factor (LN). The weight on the Treasury bond is constrained to lie between -1.0 and 8.0. Buy and hold corresponds to the strategy of investors passively holding Treasury bonds.

Lopez-Salido et al. (2017) show that credit market sentiment has predictive power for macroeconomic conditions. Moreover, Ludvigson and Ng (2009, 2011) find that macroeconomic variables are relevant to bond risk premia. Therefore, we examine whether TSENT can also predict future macroeconomic performance. Specifically, we estimate the following predictive regression:

$$\Delta Econ_{t+h} = a_0 + a_1 TSENT_t + a_2 CP_t + a_3 LN_t + a_4 FB_t + e_{t+1}, \quad (13)$$

where $\Delta Econ_{t+h} = \frac{1200}{h+1} \ln\left(\frac{Econ_{t+h}}{Econ_{t-1}}\right)$. $\Delta Econ_{t+h}$ denotes the monthly growth rate of an economic indicator in a forecasting horizon with h months (scaling by a constant).¹³ We are interested in the 1-, 3- and 12-month-ahead forecast horizons. The timing of the model is due to the fact that the value of $Econ_t$ is not readily available to investors in month t because of possible reporting lags. Following Gilchrist and Zakrajšek (2012), we choose payroll employment, civilian unemployment rate, and industrial production as our candidate economic variables.¹⁴ For the unemployment rate only, $\Delta Econ_{t+h} = \frac{Econ_{t+h} - Econ_{t-1}}{h+1}$.

Panel A of Table 8 shows that TSENT can forecast the growth in payroll employment at all three horizons in both univariate and multivariate settings. In the multivariate setting, results show that FB is related to payroll employment changes over the next 1 and 3 months. But the forecasting power vanishes at the 12-month horizon. In addition, the magnitude of the coefficient on FB is smaller than that on TSENT. Meanwhile, LN and CP show no predictive power for future payroll employment growth. In Panels B and C, we report the results of changes in the unemployment rate and industrial production, respectively. Similar to Panel A, TSENT continues to display the strongest predictive power for future economic conditions. Except for TSENT, only FB shows some limited forecasting power for unemployment rate changes.

More important, the coefficients on TSENT in all panels have the correct signs and are consistent with the information-relevant story. Our results show that the optimism of Treasury institutional investors is associated with the deterioration of the overall economy (lower payroll employment, higher unemployment rate, and lower industrial production) over the next few months. Because of the well-known flight-to-quality phenomenon, Treasury bonds are preferred in the market, which leads to the positive relation between TSENT and future Treasury bond returns. In other words, investors tend to allocate their capital to safe assets and boost Treasury bond returns when the economy is worsening. The monetary policy literature also suggests that the Federal Reserve is inclined to a low-interest-rate policy during bad times.

6 | CONCLUSION

We find that investor sentiment plays a significant role in the Treasury market. Our Treasury sentiment proxy, TSENT, is a statistically and economically significant predictor of future Treasury bond excess returns with maturities ranging from 2 through 5 years. TSENT consistently exhibits stronger in-sample predictive power than FB, CP, and LN, which are conventional predictors used in the literature. Moreover, we find that TSENT provides substantial out-of-sample forecasting gains relative to baseline models such as the prevailing mean benchmark model or an AR(1) model. We also show that TSENT generates substantial utility gains for a mean-variance investor and provides high Sharpe ratios, which validates the economic value of TSENT's predictive ability. Finally, the predictive power of TSENT appears to stem from its forecasting power for macroeconomic performance.

¹³To save space, we report results only for 2-year Fama–Bliss forward spreads (FB2). The findings based on Equation (13) are similar when longer horizon Fama–Bliss forward spreads are used.

¹⁴In addition to the three economic indicators, we investigate the Consumer Price Index (CPI) and find that TSENT negatively predicts inflation changes over the next 12 months. The results are available from the authors upon request.

TABLE 8 Future economic activity forecasts.

	<i>h</i> = 1 month		<i>h</i> = 3 months		<i>h</i> = 12 months	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Payroll employment</i>						
TSENT	-0.670*** (0.191)	-0.737*** (0.121)	-0.723*** (0.201)	-0.762*** (0.135)	-0.815*** (0.190)	-0.768*** (0.156)
CP		-0.243 (0.232)		-0.260 (0.241)		-0.156 (0.218)
LN		-0.431 (0.271)		-0.390 (0.284)		-0.428 (0.273)
FB		-0.477*** (0.175)		-0.399** (0.175)		-0.165 (0.190)
R ² (%)	13.510	31.319	16.968	32.843		36.729
<i>Panel B: Unemployment rate</i>						
TSENT	0.042*** (0.011)	0.050*** (0.009)	0.040*** (0.013)	0.044*** (0.009)	0.032** (0.013)	0.037*** (0.009)
CP		0.002 (0.017)		0.009 (0.017)		-0.001 (0.014)
LN		0.017 (0.018)		0.023 (0.018)		0.024 (0.016)
FB		0.028* (0.015)		0.024 (0.015)		0.024 (0.017)
R ² (%)	6.611	9.801	7.445	14.146	8.956	17.992
<i>Panel C: Industrial production</i>						
TSENT	-1.312** (0.613)	-1.170* (0.635)	-1.268** (0.619)	-0.977* (0.588)	-0.897* (0.509)	-0.665 (0.492)
CP		0.246 (0.842)		0.018 (0.841)		0.255 (0.595)
LN		-0.663 (1.221)		-0.860 (1.014)		-0.860 (0.734)
FB		0.252 (0.730)		0.505 (0.716)		0.515 (0.684)
R ² (%)	4.448	5.861	5.764	9.232	4.868	11.638
<i>Panel D: CPI</i>						
TSENT	0.136 (0.324)	0.430 (0.270)	0.096 (0.303)	0.352 (0.251)	0.084 (0.173)	0.337** (0.165)

(Continues)

TABLE 8 (Continued)

	<i>h</i> = 1 month		<i>h</i> = 3 months		<i>h</i> = 12 months	
	(1)	(2)	(3)	(4)	(5)	(6)
CP		−0.568		−0.408*		−0.340***
		(0.393)		(0.231)		(0.113)
LN		−0.036		0.061		−0.077
		0.191		(0.139)		(0.088)
FB		0.599		0.594		0.633***
		(0.458)		(0.361)		(0.165)
<i>R</i> ² (%)	0.194	3.551	0.180	4.835	0.529	20.996

Note: This table reports the macroeconomic conditions predictive regression results. The predictors include SENTIX Treasury market sentiment (TSENT), Fama–Bliss forward spreads (2, 3, 4, and 5 years) (FB), Cochrane–Piazzesi (2005) forward rate factor (CP), and Ludvigson–Ng (2011) factor (LN). All predictors are standardized using their respective sample means and sample standard deviations. In Panels A, B, C and D, the dependent variables are the change in payroll employment, the change in unemployment, the change in industrial production and the change in the Consumer Price Index (CPI), respectively. *h* is the forecast horizon. The regressions are estimated using the Newey–West heteroskedastic and autocorrelation robust covariance matrix. Standard errors are provided in parentheses. The sample period is February 2001 to December 2019.

**p* < 0.10.

***p* < 0.05.

****p* < 0.01.

Our findings provide important implications for future studies. We show that not all investor sentiments are irrational and that TSENT contains valuable information about future Treasury bond excess returns. Future research should explore the role of investor sentiment in other asset markets.

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